

Compositionality, Negation and Modality

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Team Semantics: Linguistic and Philosophical Applications
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Outline

1. Compositionality

2. Negation

3. Modality

4. Definability

Semantic type

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- Two ways to think about compositionality.

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- ▶ For total semantics, or when the interpreted domain is closed under subterms, the two formulations coincide.
- ▶ The hard question is: **which values make them true without trivializing meaning?**

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Two expressions have the same **Fregean value** when they fit into the same interpreted contexts and no such context distinguishes them:

$$e \equiv_{\mu}^F f \iff e \sim_{\mu} f \text{ and } \forall \pi[-] (\pi[e] \in X \Rightarrow \mu(\pi[e]) = \mu(\pi[f]))$$

The **Fregean value** of e is its complete observable behavior:

$$|e|_{\mu}^F = \{f : f \equiv_{\mu}^F e\}.$$

Hodges's Lifting Lemma

If every expression occurs inside some expression in X , then $e \mapsto |e|_\mu^F$ is a total, compositional semantics.

$$|e|_\mu^F = |f|_\mu^F \iff \text{no interpreted context distinguishes } e \text{ and } f.$$

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The **challenge** is to find denotations rich enough to recover inherited strategic information.

Singleton truth and uniformity

Let M have two distinct elements a, b , and consider

$$\psi(x) := (\exists y / \{x\}) x = y.$$

	x
i_0	a
i_1	b

- ▶ On each singleton team, Eloise can choose $y = i(x)$.
 - ▶ On $X = \{i_0, i_1\}$, she must choose y uniformly, independently of x .
 - ▶ No single choice works for both a and b .
-
- ▶ $M, \{i_0\} \models \psi$
 - ▶ $M, \{i_1\} \models \psi$
 - ▶ But $M, \{i_0, i_1\} \not\models \psi$
 - ▶ As we have seen, truth on all points does not determine truth on the team.

Universal lifting

Suppose open formulas had only ordinary Tarskian extensions

$$\llbracket \varphi \rrbracket^M \subseteq M^{\text{FV}(\varphi)}$$

and team support were obtained by universal lifting:

$$M, X \models^+ \varphi \iff X \subseteq \llbracket \varphi \rrbracket^M$$

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Every formula would be *flat*. But $\psi(x) = (\exists y/\{x\})x = y$ is supported by both singleton subteams of $\{i_0, i_1\}$ and not by their union.

Arbitrary sets of assignments

The flatness argument rules out one lifting recipe. Cameron and Hodges prove a stronger result. There is no *total compositional* semantics for all IF formulas that

1. agrees with game-theoretic truth on IF sentences, and
2. assigns every open formula merely a set of assignments.

For every $n \geq 2$, some n -element model requires more than 2^n distinct values for formulas $\varphi(x)$, while subsets of $M^{\{x\}}$ supply only 2^n candidates.

Shifting the parameter

$$t_M(\varphi, X) = 1 \text{ iff } M, X \models \varphi.$$

For split disjunction, $t_M(\varphi \vee \psi, X) = 1 \iff \exists Y, Z (X = Y \cup Z \ \& \ t_M(\varphi, Y) = 1 \ \& \ t_M(\psi, Z) = 1)$

The parameter X is **shifted**: the constituents are evaluated at Y and Z , not at the original X . Hence truth-at- X is not, by itself, a compositional value.

Instead of assigning φ a truth value at one team, we assign it its full truth profile: the function mapping each team to 1 or 0, equivalently **the set of all teams** that support it.

$$\widehat{t}_M(\varphi) := \lambda X. t_M(\varphi, X) \quad \cong \quad \{X \mid M, X \models \varphi\}$$

Teams

For a variable domain V , let $\text{Teams}_M(V) = \mathcal{P}(M^V)$. Hodges's positive denotation is the collection of the supporting teams:

$$\llbracket \varphi \rrbracket_M^+ = \{X \in \text{Teams}_M(\text{FV}(\varphi)) \mid M, X \models^+ \varphi\}$$

Algebra on team properties

Fix V . Let $P, Q \subseteq \text{Teams}_M(V)$. For $E_{y/W}$ let $P \subseteq \text{Teams}_M(V \cup \{y\})$.

$$P \cap Q = \{X \mid X \in P \text{ and } X \in Q\},$$

$$P \oplus Q = \{X \mid \exists Y \in P \exists Z \in Q (X = Y \cup Z)\},$$

$$E_{y/W}(P) = \{X \mid \exists W\text{-uniform } F : X \rightarrow M \ X[F/y] \in P\}$$

Here F is W -uniform iff

$$i \upharpoonright (\text{Dom}(X) \setminus W) = j \upharpoonright (\text{Dom}(X) \setminus W) \quad \Rightarrow \quad F(i) = F(j)$$

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Then the positive clauses become genuine operations on denotations:

$$\llbracket \varphi \wedge \psi \rrbracket_M^+ = \llbracket \varphi \rrbracket_M^+ \cap \llbracket \psi \rrbracket_M^+$$

$$\llbracket \varphi \vee \psi \rrbracket_M^+ = \llbracket \varphi \rrbracket_M^+ \oplus \llbracket \psi \rrbracket_M^+$$

$$\llbracket (\exists y/W) \varphi \rrbracket_M^+ = E_{y/W}(\llbracket \varphi \rrbracket_M^+)$$

Dual negation forces bilateral meanings

A positive family alone does not in general determine the positive family of the dual negation

$$\mu_M(\varphi) = \langle P_\varphi, N_\varphi \rangle = \langle \{X \mid M, X \models^+ \varphi\}, \{X \mid M, X \models^- \varphi\} \rangle$$

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- ▶ ‘Classical’ negation $\neg_c$: only allow negation to occur in front of formulas in the classical fragment of the language (or only in front of propositional variables):

$$s \models \neg_c \alpha \iff \{v\} \not\models \alpha \text{ for all } v \in s$$

Boolean negation $\sim$

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- ▶ Conjunction and negation form a functionally complete set of connectives. With $\sim$ and $\wedge$, we therefore have classical propositional logic over teams.
- ▶ Addition of $\sim$ to a logic will typically break all closure properties and lead to a great increase in expressive power. E.g., classical propositional logic extended with $\sim$ can express all sets of teams; dependence logic with $\sim$ is expressively equivalent with full second-order logic.

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- ▶ How should $\|\phi\|^+$ and $\|\phi\|^-$ relate to one another?

Intuitionistic negation $\neg_i$

$$s \models \neg_i \phi \iff \forall t \subseteq s: \text{ if } t \models \phi, \text{ then } t = \emptyset$$

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- ▶ Implicit quantification over teams can allow for great expressive power. E.g., while dependence logic is equivalent to existential second-order logic (over sentences), its extension with $\rightarrow$ is equivalent to full second-order logic.

Dual negation: extreme failure of determination

With the dual negation preservation of equivalents under replacement does not hold in the scope of negation:

$$\neg \forall x (\exists y / \{x\}) x = y \equiv \perp \quad \text{but} \quad \neg \neg \forall x (\exists y / \{x\}) x = y \equiv \forall x (\exists y / \{x\}) x = y \neq \perp$$

$$\neg ne \equiv \perp \quad \text{but} \quad \neg \neg ne \equiv ne \neq \top \equiv \neg \perp.$$

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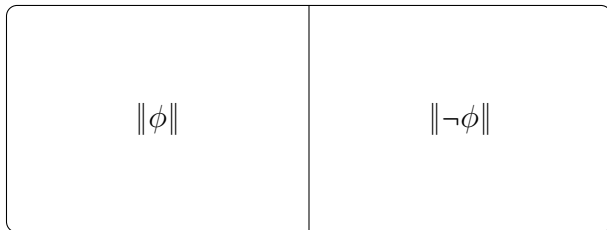
$$\|\neg ne\|^+ = \|\perp\|^+ \quad \text{but} \quad \|\neg ne\|^- \neq \|\perp\|^-.$$

Burgess showed that for IF sentences, this failure of determination is extreme: for an IF sentence, given only $\|\phi\|^+ = \{\mathcal{M} \mid \mathcal{M} \models \phi\}$, $\|\phi\|^- = \{\mathcal{M} \mid \mathcal{M} \models \neg \phi\}$ can be any class of models disjoint from $\|\phi\|^+$ that is expressible in IF.

Burgess' Theorem

In classical first-order logic, $\mathcal{M} \models \neg\phi \iff \mathcal{M} \not\models \phi$, so given

$\|\phi\| = \{\mathcal{M} \mid \mathcal{M} \models \phi\}$, to find $\|\neg\phi\|$, simply take the complement of $\|\phi\|$.

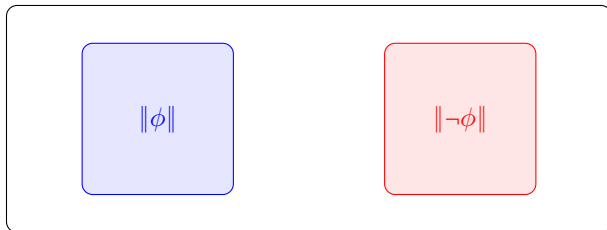


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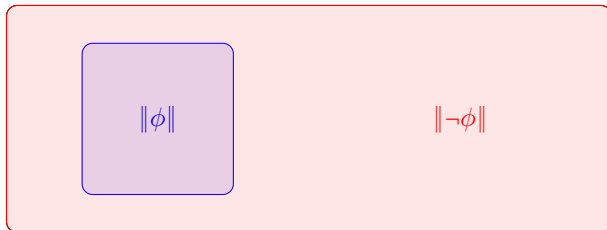


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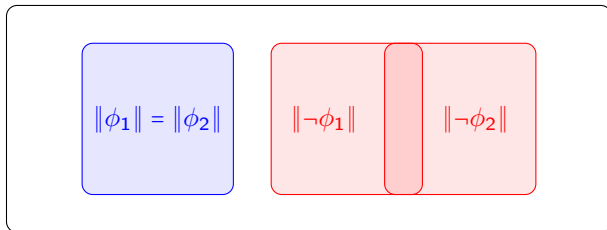


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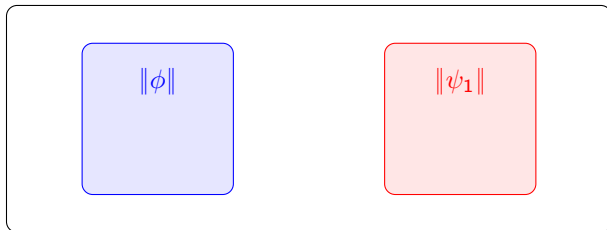
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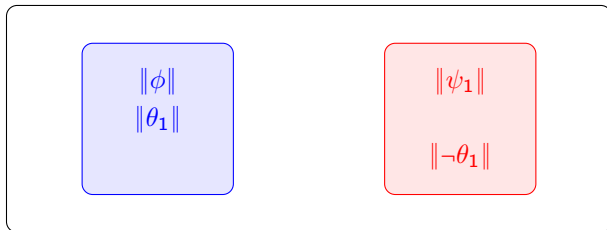
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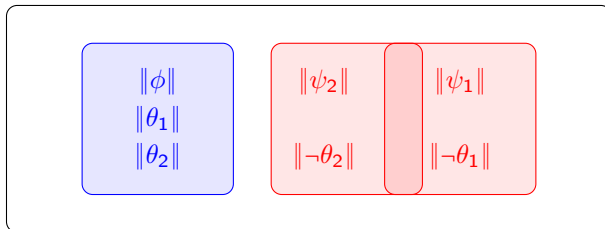
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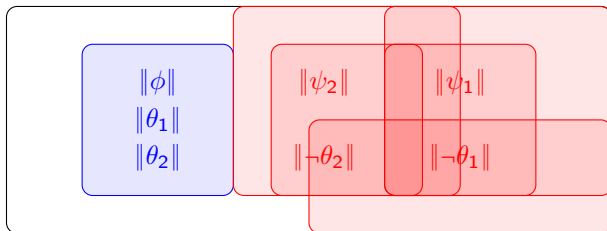
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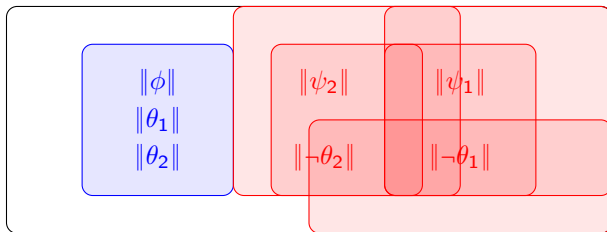
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So if we only know $\|\phi\|$, $\|\neg\phi\|$ can be any class of models $\mathcal{P}$, as long as that class is IF-definable ($\mathcal{P} = \|\psi\|$ for some ψ) and disjoint with $\|\phi\|$.

Assume all models are of size ≥ 2 .

Lemma: There is θ_0 s.t. $\theta_0 \equiv \perp \equiv \neg\theta_0$. (E.g. the matching pennies sentence $\forall x(\exists y/\{x\})x = y$.)

Assume all models are of size ≥ 2 .

Lemma: There is θ_0 s.t. $\theta_0 \equiv \perp \equiv \neg\theta_0$. (E.g. the matching pennies sentence $\forall x(\exists y/\{x\})x = y$.)

Separation Theorem: If $\|\phi\|$ and $\|\psi\|$ are disjoint, there is an η s.t. $\phi \models \eta$ and $\psi \models \neg\eta$. (Follows from ESO-equivalence and Craig's interpolation for classical first-order logic.)

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Burgess' intended his theorem to serve in part as a point against IF and Hintikka's philosophical ambitions:

In recent years Hintikka and co-workers have revived a variant version of the logic of Henkin sentences under the label “independence-friendly” logic, have restated many theorems about existential second-order sentences for this “new” logic, and have made very large claims about the philosophical importance of the theorems thus restated. In discussion, pro and con, of such philosophical claims it has not been sufficiently emphasized that contrariety [dual negation], the only kind of “negation” available, fails to correspond to any operation on classes of models. For this reason it seemed worthwhile to set down, in the form of the corollary above, a clear statement of just how total the failure is.

With bilateral meanings $(\|\phi\|^+, \|\phi\|^-)$, and assuming double negation elimination, the negation corresponds simply to the operation flipping the positive and negative denotations:

$$\neg : (\|\phi\|^+, \|\phi\|^-) \mapsto (\|\phi\|^-, \|\phi\|^+).$$

Expressive completeness

A logic L is expressively complete for some class $\mathbb{P}$ if

$$\mathbb{P} = \llbracket L \rrbracket, \quad \text{where } \llbracket L \rrbracket = \{\llbracket \phi \rrbracket \mid \phi \in L\}.$$

We use $\llbracket \phi \rrbracket$ to refer to the set of semantic objects satisfying ϕ :

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(for sentences ϕ , this can be simplified to $\{\mathcal{M} \mid \mathcal{M} \models \phi\}$).

Some expressive completeness results

- Over sentences, dependence logic and IF logic are both expressively complete for the class of ESO-definable sets of FO structures.

Intuitively, expressive completeness of e.g. propositional dependence logic says that not only are all meanings of dependence logic downward- and empty-team-closed, but propositional dependence logic can capture all such meanings.

Väänänen 2007, Kontinen & Väänänen 2009, Ciardelli & Roelofsen 2011, Yang & Väänänen 2016, Anttila & Knudstorp 2025

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- ▶ BSML is expressively complete for the class of convex and union-closed sets of pointed Kripke models.

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A logic is bicomplete for a class of pairs $\mathbb{P}$ if $\|L\|^\pm = \mathbb{P} \cap (\|L\| \times \|L\|)$.

Bicompleteness for IF sentences

Burgess' theorem, together with its converse (which implies: $\mathcal{M} \models \phi$ and $\mathcal{M} \models \neg\phi$ are never true simultaneously— $\|\phi\|^+ \cap \|\phi\|^- = \emptyset$) shows that with respect to sentences, IF is bicomplete for disjoint pairs.

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The positive expressive completeness theorems IF says that IF can express all and only meanings $\|\phi\|$ that are ESO-expressible, downward- and empty-team closed.

The bicompleteness theorem says that among the IF-expressible pairs, the ones corresponding to a bilateral meaning $\|\phi\|^\pm$ are precisely the disjoint ones. In other words, the property of pairs captured by the IF dual negation is disjointness.

Bicompleteness for IF formulas

Say that formulas ϕ and ψ are $\perp$ -incompatible if $X \models \phi$ and $X \models \psi$ together imply $X = \emptyset$. I.e., $\phi, \psi \models \perp$.

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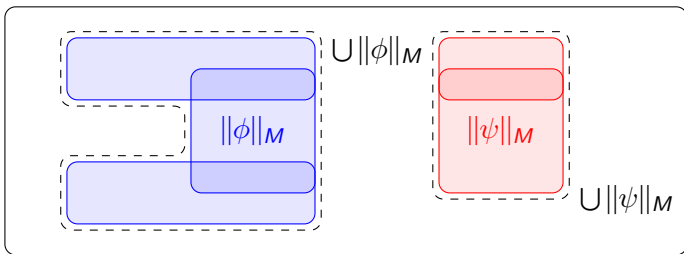
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BSML is bicomplete for ground-incompatible pairs.



Ground-incompatibility and $\perp$ -incompatibility

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In a downward-closed setting, the two are equivalent: for $\perp$ -incompatible downward-closed ϕ and ψ , if $X \models \phi$ and $Y \models \psi$, then $X \cap Y \models \phi \wedge \psi$, so $X \cap Y = \emptyset$.

Ground-incompatibility in information states

Interpret teams as information states and extend classical logic with the epistemic 'might'-operator:

$$X \models \blacklozenge \phi \iff \exists Y \subseteq X : Y \models \phi \text{ and } Y \neq \emptyset$$

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If ϕ and ψ are $\perp$ -incompatible but not ground-incompatible, it is not only factual information but also information regarding epistemic uncertainty that makes them incompatible. E.g., epistemic contradictions:

“It is not raining but it might be raining”

$$\neg r, \blacklozenge r \models \perp \qquad |\neg r \wedge \blacklozenge r|_M \neq \emptyset.$$

More incompatibility notions/pair properties

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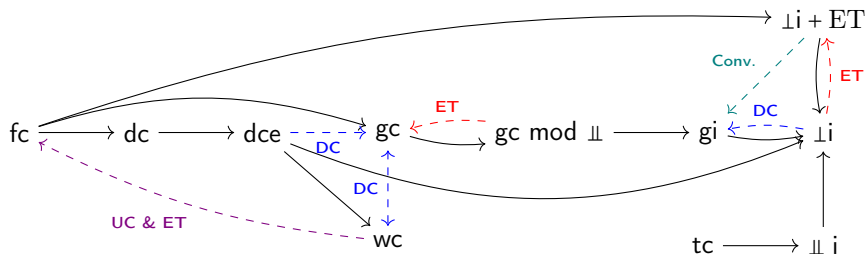
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Propositional dependence logic with the dual negation is bicomplete for ground-complementary pairs.

Pair property	Definition(s)	Bicomplete logics
$\perp$ -incompatible ($\perp i$)	• $[s \models \phi_0 \text{ and } s \models \phi_1] \implies s = \emptyset$ • $\phi_0, \phi_1 \models \perp$ • $\perp i + ET$ or $\perp i$	D, IF
Ground-incompatible (gi)	• $[s \models \phi_0 \text{ and } t \models \phi_1] \implies s \cap t = \emptyset$ • $\phi_0 \cap \phi_1 = \emptyset$	D, IF, BSML, BSML($\sqcup$), PL(ne, $\sqcup$)
$\sqcup$ -incompatible ($\sqcup i$)	• ϕ_0 and ϕ_1 never jointly true • $\phi_0, \phi_1 \models \sqcup$ • $\ \phi_0\ \cap \ \phi_1\ = \emptyset = \ \sqcup\$	
$\perp$ -incompatible and empty team prop. ($\perp i + NE$)	• $[s \models \phi_0 \text{ and } s \models \phi_1] \iff s = \emptyset$ • $\phi_0, \phi_1 \models \perp$ and $\phi_0, \phi_1 \not\models \sqcup$ • $\ \phi_0\ \cap \ \phi_1\ = \{\emptyset\} = \ \perp\$	D, IF
World-complementary (wc)	• $\{w\} \models \phi_0 \iff \{w\} \not\models \phi_1$	PL($\neg(\cdot)$), PL
Team-complementary (tc)	• $s \models \phi_i \iff s \not\models \phi_{1-i}$ • $\ \phi_i\ = \ \top\ \setminus \ \phi_{1-i}\$	PL($\sim$) (w.r.t. $\sim$)
Flat-complementary (fc)	• wc and ϕ_0, ϕ_1 flat • $\ \phi_i\ = \wp(\top \setminus \phi_{1-i})$ • $\ \phi_i\ = \{s \mid t \models \phi_{1-i} \implies s \cap t = \emptyset\}$ • $\ \phi_i\ = \bigcup \{P \subseteq \ \top\ \mid P, \ \phi_{1-i}\ \text{ G-I}\}$	PL
ϕ_1 down-set complement (dc) of ϕ_0	• $s \models \phi_1 \iff [[t \models \phi_0 \text{ and } t \subseteq s] \implies t = \emptyset]$	InqB (w.r.t. $\neg_i$), PL
Down-set complements (on either side) (dce)	• ϕ_1 dc of ϕ_0 or ϕ_0 dc of ϕ_1	HS, PL
Ground-complementary (gc)	• $\phi_i = \top \setminus \phi_{1-i}$	PL($\neg(\cdot)$), PL
Ground-complementary mod $\sqcup$	• $\phi_i = \top \setminus \phi_{1-i}$ or $\phi_0 \equiv \sqcup$ or $\phi_1 \equiv \sqcup$	PL(ne), PL($\neg(\cdot)$), PL
All pairs		PL(ne*, $\sqcup$)

Relations between incompatibility notions



Bicompleteness and indetermination

We can think of bicompleteness theorems with respect to different incompatibility notions as measuring the extent to which the positive and negative meanings are independent, with a stronger notion corresponding to less independence:

- Classical logic interpreted on teams is bicomplete for pairs $(\mathcal{P}, \mathcal{Q})$ where $\mathcal{P}$ and $\mathcal{Q}$ are both flat, and $\mathcal{P} = \|\top\| \setminus \mathcal{Q}$. $\|\phi\|$ completely determines $\|\neg\phi\|$, and vice versa.

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- ▶ If a logic is bicomplete for $\perp$ -incompatible pairs but not for ground-complementary pairs, we do not even know that, but we know $\|\phi\| \cap \|\neg\phi\| \subseteq \{\emptyset\}$.

Outline

1. Compositionality
2. Negation
3. **Modality**
4. Definability

Modality over information states

In ordinary Kripke semantics, formulas denote sets of worlds $\llbracket \varphi \rrbracket_M \subseteq W$ and a modal operator transforms one proposition into another:

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In team semantics, formulas are evaluated at information states $T \subseteq W$. Their denotations are therefore **sets of states**:

$$\llbracket \varphi \rrbracket_M = \{ T \subseteq W : M, T \models \varphi \} \subseteq \mathcal{P}(W)$$

Hence a modal operator acts at the higher semantic type

$$\mathcal{P}(\mathcal{P}(W)) \longrightarrow \mathcal{P}(\mathcal{P}(W))$$

Modality over information states

In ordinary Kripke semantics, formulas denote sets of worlds $\llbracket \varphi \rrbracket_M \subseteq W$ and a modal operator transforms one proposition into another:

$$\mathcal{P}(W) \longrightarrow \mathcal{P}(W)$$

In team semantics, formulas are evaluated at information states $T \subseteq W$. Their denotations are therefore **sets of states**:

$$\llbracket \varphi \rrbracket_M = \{ T \subseteq W : M, T \models \varphi \} \subseteq \mathcal{P}(W)$$

Hence a modal operator acts at the higher semantic type

$$\mathcal{P}(\mathcal{P}(W)) \longrightarrow \mathcal{P}(\mathcal{P}(W))$$

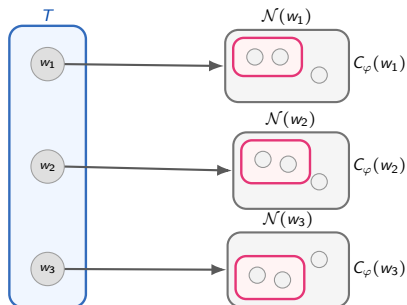
- ▶ Starting from an input state T , the modality determines one or more admissible successor states.
- ▶ It then specifies how the complement is evaluated over those successors.
- ▶ Variety of modalities: different choices yield genuinely different modal operators.

Some independent choices

1. Should modality move to one **collective successor state** or many local ones?
2. Should the model associate worlds with **worlds** or with **states**?
3. Should the modal complement see only pointwise truth, or also **relations among possibilities**?

Local modalities

- ▶ A local modality does not build one joint successor team for the whole input team.
- ▶ For each source world $w \in T \subseteq W$, look at successor information states available at w .
- ▶ The complement is evaluated **locally**, at successor states attached to each single w .
- ▶ Local modalities have the form: $M, T \models O\varphi \iff \forall w \in T \ C_\varphi(w)$



General local-state models

- ▶ Let $M = (W, V, \mathcal{N})$, where $\mathcal{N}(w) \subseteq \mathcal{P}(W)$ is the family of admissible successor states at w .
- ▶ Usually we require successor states to be nonempty.
- ▶ For a Kripke frame (W, R) , the most common choice is:

$$\mathcal{N}_R(w) = \mathcal{P}^+(R[w]) = \{s \subseteq R[w] : s \neq \emptyset\}$$

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- ▶ The pooled successor range is:

$$\rho(w) = \bigcup \mathcal{N}(w)$$

- ▶ If $\mathcal{N}(w) = \mathcal{P}^+(R[w])$, then:

$$\rho(w) = R[w]$$

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- ▶ If $\mathcal{N}(w) = \mathcal{P}^+(R[w])$, then:

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- ▶ The local architecture separates two questions:
 - ▶ which source worlds $w \in T$ must be checked?
 - ▶ which successor states $S \in \mathcal{N}(w)$ must support the complement?

Three basic local modalities

Fix a state formula φ .

- **Existential-state modality:**

$$M, T \models \Diamond_{\mathcal{N}}\varphi \iff \forall w \in T \exists S \in \mathcal{N}(w) M, S \models \varphi$$

At every live source world, some admissible successor state supports φ .

- **Universal-state modality:**

$$M, T \models \Box_{\mathcal{N}}\varphi \iff \forall w \in T \forall S \in \mathcal{N}(w) M, S \models \varphi$$

At every source world, every admissible successor state supports φ .

- **Pooled-state modality:**

$$M, T \models \Box_{\cup}\varphi \iff \forall w \in T M, \rho(w) \models \varphi$$

At every source world, the total pooled successor range supports φ .

The local quantifier square

The source-world quantifier and the successor-state quantifier can be varied independently.

- ▶ Universal source, existential state ($\forall\exists$):

$$\forall w \in T \exists S \in \mathcal{N}(w) S \models \varphi$$

- ▶ Universal source, universal state ($\forall\forall$):

$$\forall w \in T \forall S \in \mathcal{N}(w) S \models \varphi$$

- ▶ Existential source, existential state ($\exists\exists$):

$$\exists w \in T \exists S \in \mathcal{N}(w) S \models \varphi$$

- ▶ Existential source, universal state ($\exists\forall$):

$$\exists w \in T \forall S \in \mathcal{N}(w) S \models \varphi$$

- ▶ The first two are the standard local modalities.
- ▶ The last two say that **some live possibility** has the relevant modal property, with the latter requiring that this live source possibility is robustly safe.

The local quantifier square

Existential state

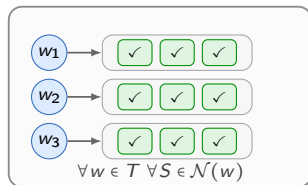
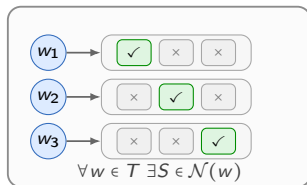
$$\exists S \in \mathcal{N}(w)$$

Universal state

$$\forall S \in \mathcal{N}(w)$$

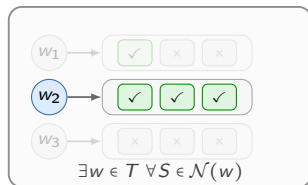
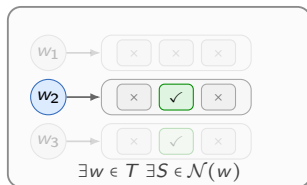
Universal source

$$\forall w \in T$$



Existential source

$$\exists w \in T$$



Universal-source modalities are flat at the input

- ▶ Let $O \in \{\Diamond_{\mathcal{N}}, \boxplus_{\mathcal{N}}, \Box_{\cup}\}$
- ▶ Each of these modalities is lifted universally over the input team:

$$M, T \models O\varphi \iff \forall w \in T \ M, \{w\} \models O\varphi$$

- ▶ Hence $O\varphi$ is flat with respect to the source team T .
- ▶ This remains true even when φ itself is non-flat.

Universal-source modalities are flat at the input

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- ▶ Hence $O\varphi$ is flat with respect to the source team T .
- ▶ This remains true even when φ itself is non-flat.
- ▶ Local modality therefore has a **reset effect**:

team-sensitive complement $\mapsto$ flat modal formula at the source level

- ▶ They are also monotone in their complement:

$$\text{If } \varphi \models \psi, \text{ then } O\varphi \models O\psi$$

Universal-state versus pooled-state

$$M, T \models \boxplus_{\mathcal{N}} \varphi \iff \forall w \in T \forall S \in \mathcal{N}(w) S \models \varphi$$

$$M, T \models \boxdot_{\cup} \varphi \iff \forall w \in T \rho(w) \models \varphi$$

Algebraically:

$$\boxplus_{\mathcal{N}} \varphi : \mathcal{N}(w) \subseteq \llbracket \varphi \rrbracket \qquad \boxdot_{\cup} \varphi : \bigcup \mathcal{N}(w) \in \llbracket \varphi \rrbracket$$

- ▶ $\boxplus_{\mathcal{N}}$ tests every admissible alternative separately.
- ▶ $\boxdot_{\cup}$ tests only the union of all alternatives.
- ▶ Thus $\boxplus_{\mathcal{N}}$ is **alternative-sensitive**.
- ▶ $\boxdot_{\cup}$ is **range-sensitive**: it forgets the internal decomposition of $\mathcal{N}(w)$.

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- ▶ If φ is union closed, then $\boxplus_{\mathcal{N}} \varphi \models \boxplus_{\cup} \varphi$.
- ▶ If φ is downward closed, then $\boxplus_{\cup} \varphi \models \boxplus_{\mathcal{N}} \varphi$.

Universal-state versus pooled-state

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- ▶ If φ is downward closed, then $\boxdot_{\cup} \varphi \models \boxplus_{\mathcal{N}} \varphi$.
- ▶ BSMML uses: $\Diamond_{\mathcal{N}_R}$ and $\boxdot_{\cup}$.
- ▶ Inquisitive modal logic uses: $\boxdot_{\cup}$ and $\boxplus_{\Sigma}$.

Ordinary modal idempotence

For ordinary propositional meanings $X \subseteq W$:

$$\Diamond_R(X) = \{w : \exists v (wRv \ \& \ v \in X)\} \quad \Box_R(X) = \{w : R[w] \subseteq X\}$$

Applying a modality twice replaces R by R^2 :

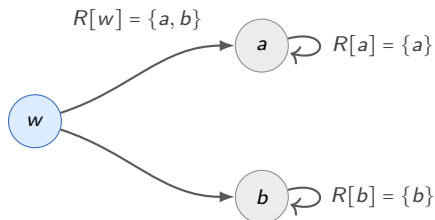
$$\Diamond_R^2(X) = \Diamond_{R^2}(X) \quad \Box_R^2(X) = \Box_{R^2}(X)$$

Thus, as identities on all propositions X :

$$\Diamond_R^2 = \Diamond_R \quad \Longleftrightarrow \quad \Box_R^2 = \Box_R \quad \Longleftrightarrow \quad R^2 = R$$

- ▶ Ordinary modality only compares the **sets of worlds** reachable in one and two steps.
- ▶ It does not ask how the two-step range is distributed among the intermediate successors.

Example



- ▶ One step: $R[w] = \{a, b\}$
- ▶ Two steps: $R^2[w] = \{a, b\}$
- ▶ The two-step range recovers exactly the one-step range.
- ▶ Recovery may be **distributed**: a preserves only a , while b preserves only b .

Families of states

For local state modalities over $\mathcal{N}_R(w) = \mathcal{P}^+(R[w])$, one step at w tests the entire family $\mathcal{P}^+(R[w])$.

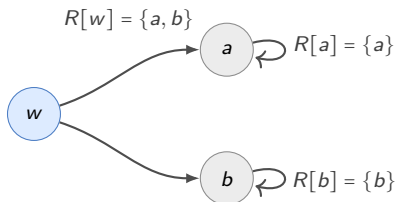
Two local steps test instead $\bigcup_{v \in R[w]} \mathcal{P}^+(R[v])$.

So idempotence on all state properties $A \subseteq \mathcal{P}(W)$ requires:

$$\mathcal{P}^+(R[w]) = \bigcup_{v \in R[w]} \mathcal{P}^+(R[v])$$

- ▶ This is stronger than $R^2[w] = R[w]$.
- ▶ It requires preservation of the **available successor states**, not merely preservation of reachable worlds.

The same frame fails locally



- ▶ One alternative-sensitive local step at w quantifies over

$$\mathcal{P}^+(R[w]) = \{\{a\}, \{b\}, \{a, b\}\}$$

- ▶ Two local steps quantify over

$$\mathcal{P}^+(R[a]) \cup \mathcal{P}^+(R[b]) = \{\{a\}, \{b\}\}$$

- ▶ Thus the collective state $\{a, b\}$ is available in one step, but disappears after the local move to a or b .

Idempotence of alternative-sensitive local modalities

The local existential-state modality $\Diamond_{\mathcal{N}_R}$ and the local universal-state modality $\Box_{\mathcal{N}_R}$ are idempotent:

$$\Diamond_{\mathcal{N}_R} \Diamond_{\mathcal{N}_R} \varphi \equiv \Diamond_{\mathcal{N}_R} \varphi \text{ and } \Box_{\mathcal{N}_R} \Box_{\mathcal{N}_R} \varphi \equiv \Box_{\mathcal{N}_R} \varphi$$

for every state formula φ exactly when every nonterminal world w satisfies:

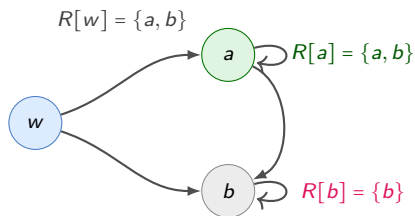
$$\forall v \in R[w] \ R[v] \subseteq R[w] \quad \text{and} \quad \exists v_0 \in R[w] \ R[v_0] = R[w]$$

- ▶ The first condition is transitivity: a further local step introduces no new possibilities.
- ▶ The second condition requires some successor v_0 to carry a **copy of the entire successor range** of w .
- ▶ Ordinary modal idempotence requires only $R^2 = R$ and therefore permits distributed recovery across different successors.
- ▶ Alternative-sensitive local idempotence requires one successor to preserve the whole range at once.

Not enough

What about $\Box_U \Box_U \varphi \equiv \Box_U \varphi$?

$$M, T \models \Box_U \varphi \iff \forall w \in T \ M, R[w] \models \varphi$$



- ▶ Suppose $\llbracket \varphi \rrbracket_M = \{\{a, b\}\}$. Then $M, \{w\} \models \Box_U \varphi$
- ▶ But $M, \{w\} \not\models \Box_U \Box_U \varphi$
- ▶ Because the second application requires both
 1. $R[a] = \{a, b\} \in \llbracket \varphi \rrbracket_M$
 2. $R[b] = \{b\} \in \llbracket \varphi \rrbracket_M$

Iterating

At a singleton $\{w\}$:

$$M, \{w\} \models \Box_U \varphi \iff R[w] \in \llbracket \varphi \rrbracket_M$$

A second application gives:

$$M, \{w\} \models \Box_U \Box_U \varphi \iff \forall v \in R[w] R[v] \in \llbracket \varphi \rrbracket_M$$

At each world w , we thus want:

$$R[w] \in \llbracket \varphi \rrbracket_M \iff \forall v \in R[w] R[v] \in \llbracket \varphi \rrbracket_M$$

for every $\llbracket \varphi \rrbracket_M \subseteq \mathcal{P}(W)$.

If $R[w] = \emptyset$, the right-hand side is vacuously true. So idempotence forces seriality

Moreover, each successor must have exactly the same range.

The frame condition is *KD45*

The condition

$$R[w] \neq \emptyset \quad \text{and} \quad wRv \Rightarrow R[v] = R[w]$$

is equivalent to seriality, transitivity, and Euclideaness.

- ▶ **Seriality:** $R[w] \neq \emptyset$
- ▶ **Transitivity:** $wRv \Rightarrow R[v] \subseteq R[w]$
- ▶ **Euclideaness:** $wRv \Rightarrow R[w] \subseteq R[v]$

Hence:

$$\Box_U \Box_U \varphi \equiv \Box_U \varphi \text{ for all } \varphi \iff R \text{ is serial, transitive, and Euclidean.}$$

What local modalities cannot express

- ▶ Universal-source local modalities have the shape:

$$M, T \models O\varphi \iff \forall w \in T \ C_\varphi(w)$$

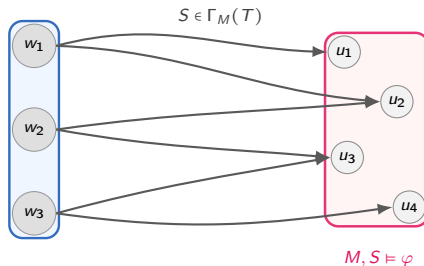
- ▶ Their strength is locality:
modal truth at T = truth at each singleton source world.
- ▶ Their limitation is the same fact: no cross-source interaction is visible.
- ▶ They cannot express coordination between successors chosen for different source worlds.
- ▶ They cannot require one joint successor team satisfying a collective property.

Global modalities

- ▶ A global modality looks at one **collective successor team**.
- ▶ Start with an information state $T \subseteq W$.
- ▶ The modal transition selects a single successor state $S \subseteq W$.
- ▶ Successors originating from different source worlds may interact inside S .
- ▶ For local modality, each $w \in T$ is instead checked separately.

$$M, T \models O\varphi \iff [\forall/\exists]S(S \in \Gamma_M(T) \ \& \ M, S \models \varphi),$$

where $\Gamma_M(T)$ is a set of admissible collective successor teams.



A general semantics

Let Γ_M assign to each team T a family of admissible successor teams:
 $\Gamma_M(T) \subseteq \mathcal{P}(W)$.

- ▶ The **angelic** global diamond is:

$$M, T \models \langle \Gamma \rangle \varphi \iff \exists S \in \Gamma_M(T) \ M, S \models \varphi.$$

- ▶ The **robust** global box is:

$$M, T \models [\Gamma] \varphi \iff \forall S \in \Gamma_M(T) \ M, S \models \varphi$$

- ▶ A deterministic box can also be defined by a distinguished successor operation F_M :

$$M, T \models \Box_F \varphi \iff M, F_M(T) \models \varphi$$

Lifting accessibility to teams

Let $M = (W, R, V)$ and let $T, S \subseteq W$.

- ▶ **Target legitimacy:**

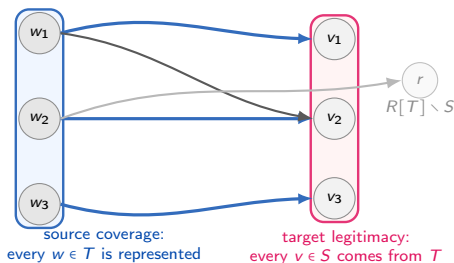
$$S \subseteq R[T]$$

Every successor world in S comes from some source world in T .

- ▶ **Source coverage:**

$$\forall w \in T \exists v \in S (wRv)$$

Every source world in T is represented in S .



Standard lax modal team semantics

- ▶ The standard lax lifting combines both:

$$TR^*S \iff S \subseteq R[T] \ \& \ \forall w \in T \ \exists v \in S \ (wRv)$$

- ▶ Equivalently, S can be obtained by choosing a nonempty $F(w) \subseteq R[w]$ for every $w \in T$ and taking $S = \bigcup_{w \in T} F(w)$.
- ▶ $M, T \models \Diamond\varphi \iff \exists S (TR^*S \ \& \ M, S \models \varphi)$
- ▶ $M, T \models \Box\varphi \iff M, R[T] \models \varphi$
- ▶ $\Diamond$ existentially selects one collective successor team.
- ▶ $\Box$ evaluates the complement at the full successor team $R[T]$.
- ▶ The two clauses are not simply existential and universal duals over the same family of successor teams.
- ▶ The strict version allows exactly one selected successor for each source world.

Dropping the source condition

- ▶ Dropping the source condition means that modality no longer requires every live possibility to have a successful continuation.
- ▶ For flat α , assuming $Y \neq \emptyset$:

$$M, X \models \Diamond_{\text{tgt}} \alpha \iff \exists w \in X \exists v (wRv \ \& \ M, v \models \alpha)$$

- ▶ Some live possibility has a successful continuation.
- ▶ Adding this modality to a downward-closed team logic introduces genuinely non-downward-closed state properties.

Dropping the target condition

- ▶ Every $w \in X$ must still have a successor in Y .
- ▶ But Y may also contain worlds that are not accessible from X .
- ▶ For downward-closed formulas, the target condition is actually redundant.
- ▶ For non-downward-closed φ , the equivalence can fail: unreachable worlds may provide additional witnesses.
- ▶ We therefore lose **successor locality**: truth may depend on worlds outside the part of the model reachable from X .

Global modality $\Box$ and idempotence

$$M, T \models \Box \varphi \iff M, R[T] \models \varphi$$

It first forms one collective successor team $R[T] = \bigcup_{w \in T} R[w]$ and then evaluates the complement once.

Applying it twice gives:

$$M, T \models \Box \Box \varphi \iff M, R^2[T] \models \varphi$$

So idempotence for all state properties requires exactly, as in standard modal logic:

$$R^2[T] = R[T] \text{ for every } T \subseteq W$$

Robust box versus full box

Assume every $w \in T$ has at least one successor, so $R[T]$ is itself admissible.

- ▶ Robust box entails full box: $[\Gamma_{\text{lax}}]\varphi \models \Box\varphi$
- ▶ Full box entails robust box when φ is downward closed: $\Box\varphi \models [\Gamma_{\text{lax}}]\varphi$
- ▶ For non-downward-closed complements, the full team may have a property that some smaller admissible successor team lacks.

What global modalities can express

Because the complement is evaluated on one joint successor team, global modalities can express collective successor properties.

- ▶ The successor team contains both p - and $\neg p$ -possibilities.
- ▶ A dependency holds across the selected successor possibilities:
 $\Diamond \text{dep}(p, q)$
- ▶ These are properties of a set of successor possibilities, not merely properties of each successor world separately.

What global modalities forget

Standard global modality remembers the output team S , but not how it was generated.

- ▶ A lax witness can be represented by: $w \mapsto F(w) \subseteq R[w]$
- ▶ Standard semantics keeps only: $S = \bigcup_{w \in T} F(w)$
- ▶ It forgets which source world produced which successor world.
- ▶ Therefore it cannot directly express properties such as:
 - ▶ every source world has two distinguishable continuations
 - ▶ two source worlds converge on the same successor
 - ▶ target facts depend on source facts
- ▶ To express those, one needs something like path-based refinements.

Bonus: modals as quantifiers

- ▶ Recall that in Lecture 4, we examined a treatment of modality as existential quantifiers over a world variable.
- ▶ So let assignments carry a current-world variable v . Accessibility can be used to bind a fresh successor variable w .
- ▶ The **relativized universal modal quantifier** is:

$$X[R(v)/w] = \{s[a/w] : s \in X \text{ and } s(v)Ra\}.$$
- ▶ $M, X \models \Box_{v \rightarrow w} \psi \iff M, X[R(v)/w] \models \psi$
- ▶ The **lax existential modal quantifier** is:

$$F : X \rightarrow \mathcal{P}^+(W), \quad \emptyset \neq F(s) \subseteq R[s(v)].$$
- ▶ $M, X \models \Diamond_{v \rightarrow w} \psi \iff \exists F \ M, X[F/w] \models \psi.$

Relation to global modalities

- ▶ Let $T = X(v)$ and let $\pi_w(Y) = Y(w)$ be the set of w -values in a team Y .
- ▶ For a lax witness F , the induced successor state is:

$$S_F = \pi_w(X[F/w]) = \bigcup_{s \in X} F(s).$$
- ▶ The possible sets S_F are exactly the standard lax successors:

$$TR^*S \iff S \subseteq R[T] \ \& \ \forall a \in T \ \exists b \in S \ (aRb)$$
- ▶ Hence, if ψ depends only on the w -projection, then

$$\Diamond_{v \rightarrow w} \psi \text{ matches } \langle \Gamma_\ell \rangle \varphi.$$
- ▶ Similarly, $\Box_{v \rightarrow w} \psi$ matches $\Box \varphi$ whenever ψ depends only on

$$\pi_w(X[R(v)/w]) = R[T].$$

Invariance

- ▶ The collapse to global modality requires the matrix to forget the source.
- ▶ Roughly, for all teams Y, Z over v, w ,
 $Y(w) = Z(w) \implies (M, Y \models \psi \iff M, Z \models \psi)$
- ▶ First-order formulas in w alone satisfy this condition.
- ▶ Dependence and variation atoms mentioning both v and w usually do not.
- ▶ Quantifier modals preserve **provenance**: which source assignment generated which successor.
- ▶ Global modalities keep only the collective successor state S .

Iteration

- ▶ Use a fresh world variable at every modal step:
 $v_0 := v, \quad v_1 := w, \quad v_2 := u, \dots$
- ▶ The relativized modal quantifiers are applied successively.
- ▶ Thus two universal steps generate a team of paths:
 $X[R(v_0)/v_1][R(v_1)/v_2] = \{s[a/v_1][b/v_2] : s(v_0)Ra \text{ and } aRb\}$
- ▶ The endpoint is v_2 , but the team still records the intermediate world v_1 and the initial world v_0 .
- ▶ Projecting to the newest variable v_i recovers endpoint-based global modal semantics. Retaining all variables yields a stronger path-sensitive semantics.

Modality need not be metaphysical

- ▶ A modal operator in team semantics can quantify over **information states**
- ▶ In Kripke-style modality, the relevant transition is usually external: wRv .
- ▶ In information-change modality, the transition may be internal to the current state: $S \subseteq T$ or $S \supseteq T$
- ▶ We have already seen this with the black diamond modality.
- ▶ Refinement corresponds to gaining information: fewer worlds remain live.
- ▶ Expansion corresponds to losing information: more worlds become live.
- ▶ We can model epistemic-like dynamic modalities.

Four order modalities

- **Refinement possibility:**

$$M, T \models \langle \downarrow \rangle \varphi \iff \exists S \subseteq T (S \neq \emptyset \ \& \ M, S \models \varphi)$$

- **Robust refinement:**

$$M, T \models [\downarrow] \varphi \iff \forall S \subseteq T \ M, S \models \varphi$$

- **Information-loss possibility:**

$$M, T \models \langle \uparrow \rangle \varphi \iff \exists S \supseteq T \ M, S \models \varphi$$

- **Robust information loss:**

$$M, T \models [\uparrow] \varphi \iff \forall S \supseteq T \ M, S \models \varphi$$

Algebra of state properties

$$P_\varphi = \{T \subseteq W : M, T \models \varphi\}$$

The order modalities transform P_φ as follows:

- Refinement possibility gives the **nonempty upward closure**:

$$\langle \downarrow \rangle P = \{T : \exists S \neq \emptyset (S \subseteq T \ \& \ S \in P)\}$$

- Robust refinement gives the **downward interior**:

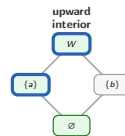
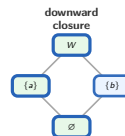
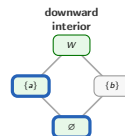
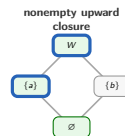
$$[\downarrow] P = \{T : \forall S \subseteq T \ S \in P\}$$

- Information-loss possibility gives the **downward closure**:

$$\langle \uparrow \rangle P = \{T : \exists S \supseteq T \ S \in P\}$$

- Robust information loss gives the **upward interior**:

$$[\uparrow] P = \{T : \forall S \supseteq T \ S \in P\}$$



An interesting fact

$$X = \mathcal{P}(W) \quad X^+ = X \setminus \{\emptyset\} \quad P \subseteq X$$

- The four order modalities act on state properties $P \subseteq X$:

$$\begin{aligned} E(P) &= \uparrow (P \cap X^+) & L(P) &= \downarrow P \\ R(P) &= \{T : \downarrow T \subseteq P\} & U(P) &= \{T : \uparrow T \subseteq P\}. \end{aligned}$$

- A nesting such as $[\uparrow]\langle\downarrow\rangle[\downarrow]\langle\uparrow\rangle\varphi$ corresponds to $\text{UERL}(P_\varphi)$.
- Although there are infinitely many syntactic nestings, many define the same transformation $\mathcal{P}(X) \rightarrow \mathcal{P}(X)$.

An interesting fact

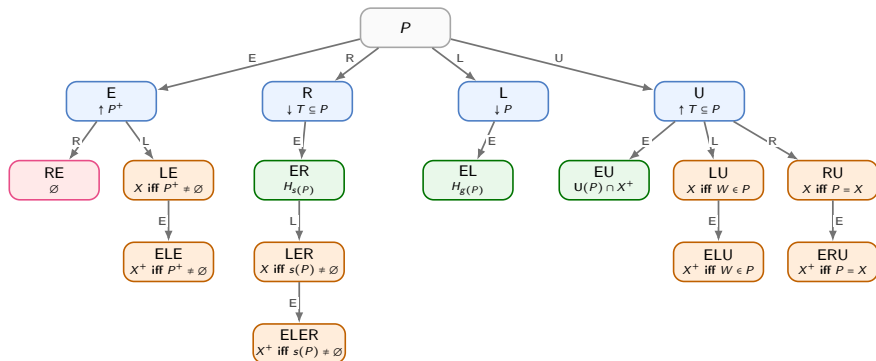
- ▶ If $|W| \geq 2$, the nonempty composites generated by E, R, L, U give exactly **sixteen** maps.
- ▶ The collapse is driven by a few invariants:
 - ▶ $g(P) = \bigcup P$: the **ground team**: the worlds occurring somewhere in P .
 - ▶ $s(P) = \{w : \emptyset \in P \text{ and } \{w\} \in P\}$: the **safe singletons**: worlds w such that both $\emptyset$ and $\{w\}$ are in P .
 - ▶ $H_A = \{T : T \cap A \neq \emptyset\}$: the hitting property generated by a set of worlds A .
- ▶ Any fixed nesting pattern stabilizes after at most two repetitions.

The sixteen maps

$$P^+ := P \cap X^+$$

$$H_A := \{T : T \cap A \neq \emptyset\}$$

$$s(P) := \{w : \emptyset \in P, \{w\} \in P\}$$



Outline

1. Compositionality
2. Negation
3. Modality
- 4. Definability**

Failure of closure under uniform substitution

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$$\models p \vee \neg p \text{ but } \not\models ne \vee \neg ne$$

This results in a disconnect between definability of formulas and definability of connectives.

Definability of formulas vs definability of contexts

Propositional inquisitive and dependence logic are expressively equivalent: each can express all downward- and empty-set closed sets of teams. Therefore, each can express all formulas of the other. However, neither can define all connectives of the other.

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Similarly, the inquisitive disjunction $\sqcup$ and the intuitionistic implication $\rightarrow$ are not definable in dependence logic.

Example: undefinability of $\rightarrow$

Assume that the syntax of dependence logic is defined as follows: the atomic formulas are propositional letters and their negations p , $\neg p$; falsum $\perp$; and dependence atoms $\text{dep}(p_1, \dots, p_n, q)$. The connectives are $\wedge$ and $\vee$.

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The implication $\rightarrow$ is not monotone: $\perp \models \top$ and $\perp \models \perp$, but $\perp \rightarrow \perp \not\models \top \rightarrow \perp$.

Uniform substitution and definability of connectives

If each atomic formulas and connective of L_1 is expressible/definable in L_2 and vice versa, the two are expressively equivalent: each can express all formulas of the other.

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Conversely, assuming closure under uniform substitution, expressive equivalence implies the definability of each connective of one logic in the other.

If L_1 can express the formula $p \star q$ of L_2 where $\star$ is a connective of L_2 , then there is a context $\theta_\star[\cdot_1, \cdot_2]$ of L_1 such that $\theta_\star[p, q] \equiv p \star q$. Then for any ϕ, ψ of L_1 , by closure under uniform substitution, also $\theta_\star[\phi, \psi] \equiv \phi \star \psi$.

Compositional translation

The connective undefinability results have the consequence that there is no compositional translation between propositional inquisitive and dependence logic, in the following sense:

A map $\tau : L_1 \rightarrow L_2$ between logics L_1 and L_2 is a compositional translation if

1. τ is meaning-preserving: for all ϕ of L_1 , $\phi \equiv \tau(\phi)$
2. τ is compositional: for each connective $\star$ of L_1 , there is a context $\theta_\star$ of L_2 such that

$$\tau(\star[\varphi_1, \dots, \varphi_n]) = \theta_\star[\tau(\varphi_1), \dots, \tau(\varphi_n)]$$

for all $\varphi_1, \dots, \varphi_n$ of L_1 .

Hierarchy of definability notions

(Under certain assumptions) each n -ary connective $\star$ defines a (partial) function $\star^{\mathbb{A}} : \mathbb{A}^n \rightarrow \mathbb{A}$, where $\mathbb{A}$ is the class of all sets of teams.

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Local definability is (roughly) the usual notion of definability of a connective.

Hierarchy of extension notions

For logics L_1 and L_2 , we say:

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- ▶ L_1 is an expressive extension of L_2 if each formula of L_2 is expressed by a formula of L_1 .

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For instance, L_1 is an expressive extension of L_2 just in case there is a translation (meaning-preserving map) $\tau : L_2 \rightarrow L_1$.

Example: convex extensions of inquisitive logic

If we find a logic that can express all convex sets of teams, that logic will be able to express all formulas of inquisitive logic InqB since these are downward closed (and hence convex).

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The logic with connectives $\rightarrow$, $\perp$, $\wedge$, and $\blacklozenge$ is expressively complete for convex sets, but one can show that $\wp$ is not definable with respect to InqB in this logic. Therefore, the logic is not a compositional extension of InqB .

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To get a convex compositional extension of inquisitive logic, we can extend instead with a variant $\wp'$ that is equivalent to $\wp$ for all InqB-formulas, and forces convexity:

$$t \models \phi \wp' \psi \iff \exists s \sqsupseteq t : s \models \phi \text{ or } s \models \psi.$$

Exercise: Bicompleteness

Consider the propositional team logic with the following syntax:

$$\phi ::= p \mid \perp \mid \neg\phi \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \wp \phi \mid \text{ne}^*$$

And the following semantics:

$$s \models p \quad : \iff \text{for all } w \in s : w(p) = 1$$

$$s \models p \quad : \iff \text{for all } w \in s : w(p) = 0$$

$$s \models \perp \quad : \iff s = \emptyset$$

$$s \models \perp \quad : \text{always the case}$$

$$s \models \text{ne}^* \quad : \iff s \neq \emptyset$$

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$$s \models \neg\phi \quad : \iff s \models \phi$$

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$$s \models \phi \wedge \psi \quad : \iff s \models \phi \text{ and } s \models \psi$$

$$s \models \phi \wedge \psi \quad : \iff \exists t, u \text{ s.t. } s = t \cup u \text{ and } t \models \phi \text{ and } u \models \psi$$

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(Continued on the next slide.)

Exercise: Bicompleteness

Denote $\phi \equiv^{\pm} \psi$ iff $\phi \equiv \psi$ and $\neg\phi \equiv \neg\psi$.

Show that:

1. There is a formula ϕ_{ne} with $\phi_{\text{ne}} \equiv^{\pm} \text{ne}$ (i.e., $t \models \phi_{\text{ne}} \iff t \neq \emptyset$ and $t \models \phi_{\text{ne}} \iff t = \emptyset$).
2. There is a formula $\phi_{\perp}$ s.t. $t \models \phi_{\perp}$ never holds, and $\neg\phi_{\perp} \equiv \top$.
3. For each ϕ , there is a ϕ^* s.t. $\phi \equiv \phi^*$, and $\neg\phi^*$ has the empty team property (hint: make use of the formula ϕ_{ne} from 1).
4. For each ϕ , there is a $\phi^{\top}$ s.t. $\phi^{\top} \equiv \phi$, and $\neg\phi^{\top} \equiv \top$ (hint: make use of the formula ϕ^* from 3).
5. This logic is bicomplete for all pairs. I.e., for all ϕ, ψ , there is a θ s.t. $\phi \equiv \theta$ and $\psi \equiv \neg\theta$. (hint: make use of the formula $\phi_{\perp}$ from 2, and the formulas $\phi_{\top}$ from 4).